Part A: Write your final answer in the blank (*each question is worth 1 mark*) **[23 marks]**

1. If $g(x) = 3x^3 - 2x$, find $g(-2)$. ______________
2. Express $\sqrt{128}$ as a mixed radical in simplest form. ______________
3. Determine the compass bearing of a boat travelling in the direction of $S30°W$. ______________
4. Simplify to a single power with positive exponents: $\frac{64x^7y^0z^{-4}}{4x^2y^8}$ ______________
5. Evaluate to 4 decimal places: $\cot 53°$ ______________
6. In which quadrant(s) can $\sec\theta = -\sqrt{2}$ exist? ______________
7. State the related acute angle for $-190°$ ______________
8. Using special angle triangles, determine the **exact** value of $\sin 300°$ ______________
9. Solve for θ if $0° \leq \theta \leq 360°$, given $\cos\theta = 0$ ______________
10. Write in the form of a power with a rational exponent: $\sqrt[3]{x^4}$ ______________
11. Evaluate as an **exact** value in rational form: $\left(\frac{64}{27}\right)^{-\frac{2}{3}}$ ______________
12. Identify each sequence as arithmetic, geometric or neither:
    a) -10, 5, -2.5, ... ______________
    b) 3, 8, 13, 18, ... ______________
13. Write the *non-recursive* general term for question 12a). ______________
14. Write the *recursive* general term for the sequence in 12b). ______________
15. For the rational expression: $\frac{6(2x-1)}{(x+4)(x+1)} \div \frac{18(2x-1)^2}{(x+4)}$
    a) Fully simplify: 15a) ______________
    b) State the restrictions: 15b) ______________
16. State the *domain* of the function using full set notation $f(x) = \sqrt{-(x+2)}$ ______________
17. State the *range* of the function using full set notation $f(x) = -(2)^x + 1$ ______________
18. In the quadratic formula, if $b^2 - 4ac$ is negative, how many roots exist? ______________
19. Determine the number of zeros for the function $f(x) = -2(x+1)^2 + 4$ ______________
20. Given the function $f(x) = \frac{1}{x-3} + 2$, state the equation of the horizontal asymptote. ______________
21. Determine the decay factor (as a decimal) for a car which depreciates 5% per year. ______________

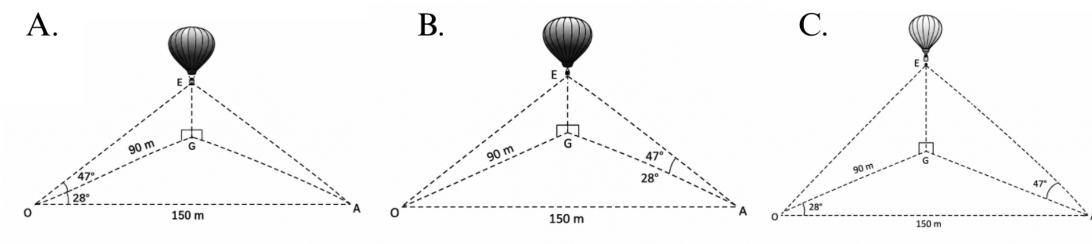
Part B: Multiple Choice and Matching

Questions 22-24 are multiple choice. (*Place the letter that indicates the best answer in the blank provided.*)

_____ 22. Solve $\tan\theta = -1$ for $0° \leq \theta \leq 360°$.

A. $45°$ or $135°$ B. $135°$ or $315°$ C. $45°$ or $315°$ D. $135°$ or $225°$

_____ 23. Olivia and Ava see their friend Emelia in a hot air balloon. Emelia is flying vertically above a point, G, on the ground. Olivia, who is standing 90m from point G, measures the angle where she is standing, between Ava and point G to be 28°. Ava measures the angle of elevation from where she is standing to the hot air balloon to be 47°. Olivia and Ava are standing 150m apart from one another. Choose the correct diagram below which would find the height of the hot air balloon.



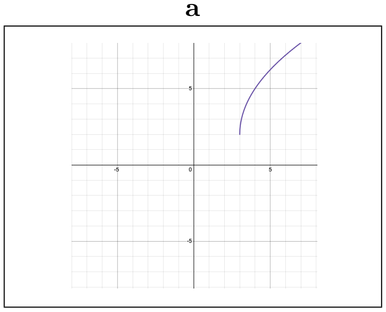
_____ 24. While getting ready to board the Ferris wheel at the Western Fair, Daniel is told by the ride operator that he will reach a maximum height of 15m during his ride. He boards the Ferris wheel at the bottom, which is reached by climbing a ramp that rises 1.5m off the ground. It takes 150 seconds for Daniel to complete one rotation. Select the *best* equation that models the height of a rider as a function of time if the model begins at the time the rider boards the wheel at the bottom.

A. $h(t) = -15\cos(150t) + 1.5$ C. $h(t) = -1.5\cos(2.4t) + 15$

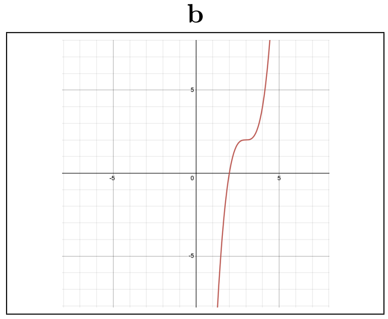
B. $h(t) = -6.75\cos[150(t - 1.5)] + 8.25$ D. $h(t) = -6.75\cos(2.4t) + 8.25$

25. Match each graph with the appropriate equation below. Write the letter of your choice in the blank.

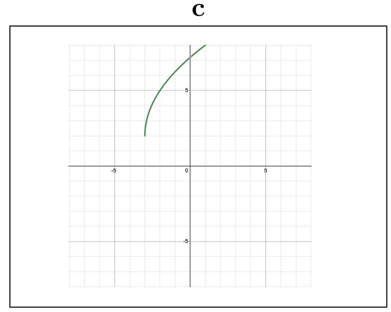
a



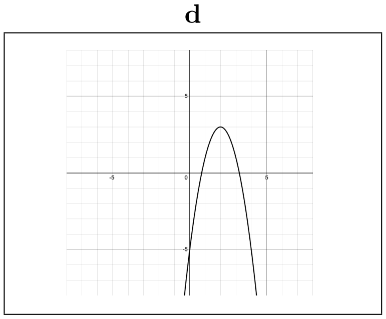
b



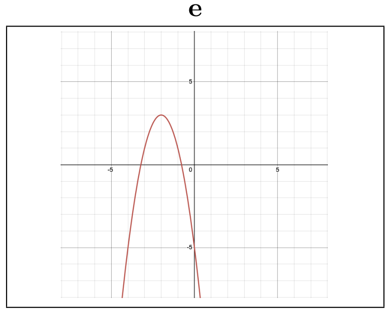
c



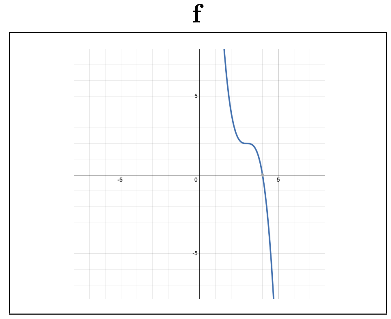
d



e



f



____ $y = 3\sqrt{x+3} + 2$ ____ $y = -2(x-2)^2 + 3$ ____ $y = -2(x-3)^3 + 2$ **[3 marks]**

____ $y = 3\sqrt{x-3} + 2$ ____ $y = -2(x+2)^2 + 3$ ____ $y = 2(x-3)^3 + 2$

Part C: Show full solutions for full marks in the spaces provided. Be sure to use proper notation and form.

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26. Simplify to a single power with the lowest base using positive exponents, then evaluate. **[5 marks]**

$$\sqrt[3]{\frac{64(x^{-1})^6}{(x)^{-3}}}$$

27. Solve for x algebraically (without using logs) in the equation below. Show all steps. **[4 marks]**

$120 = 3(6)^{\frac{x}{2}} + 12$

28. For the following problems, solve algebraically using one of the discrete formulae.

a) Determine the **simplified** general term formula for the sequence below. Then determine t_{10}. **[5 marks]**

$3, 4.75, 6.5, 8.25, \ldots, t_{10}$

b) Determine the exact value of S_{20} for the series $3, -6, 12, -24, \ldots, t_{20}$ **[4 marks]**

29. Due to habitat destruction and climate change in BC, pink salmon is depleting. Currently there are 1,000,000 salmon. The population has a half-life of ten years. Write an equation that models the salmon population as a function of time. Be sure to include let statements for your variables. **[4 marks]**

30. While standing atop a skyscraper in Manhattan, you are informed by the tour guide that the building slightly sways from its vertical during high winds. This building moves 30 cm back and forth during particularly gusty conditions. At time $t = 0$ seconds, the building was momentarily at its vertical position. The building then sways 30 cm to the left of vertical and then is back to the vertical at 30 seconds.

a) Determine the period of the sway of the skyscraper from vertical. ___________ **[1 mark]**

b) Sketch one complete cycle which models the distance travelled from vertical of the skyscraper, d, in centimetres, as a function of time, t, in seconds. Assume a sway to the right is positive. *Be sure to label your axes.* **[5 marks]**

c) Determine an equation that models the distance travelled from vertical of the skyscraper, d, as a function of time, t, in seconds. **[5 marks]**

31. Prove the following identity below: **[6 marks]**

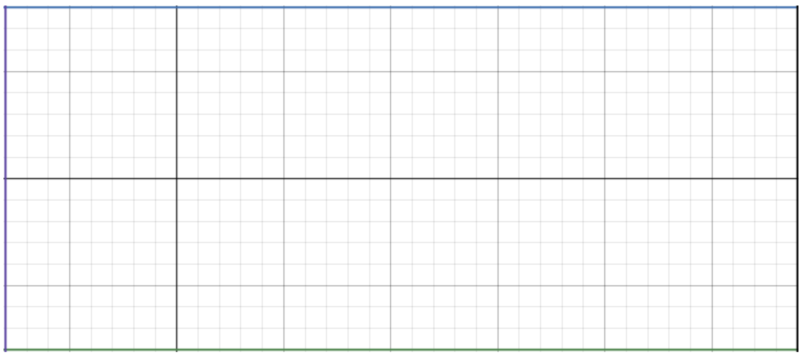
$\sec\theta - \tan\theta\sin\theta = \cos\theta$

32. a) State the characteristics of the graph given by the equation $f(x) = -3\cos[2(x - 45°)] + 1$.
Be specific about **direction** and use **proper notation/units**. **[2 marks]**

Amplitude: ________ Period: ________ Axis: ________ Phase Shift: ________

b) Graph at least one cycle using the grid below. *For full marks, you must:* **[8 marks]**

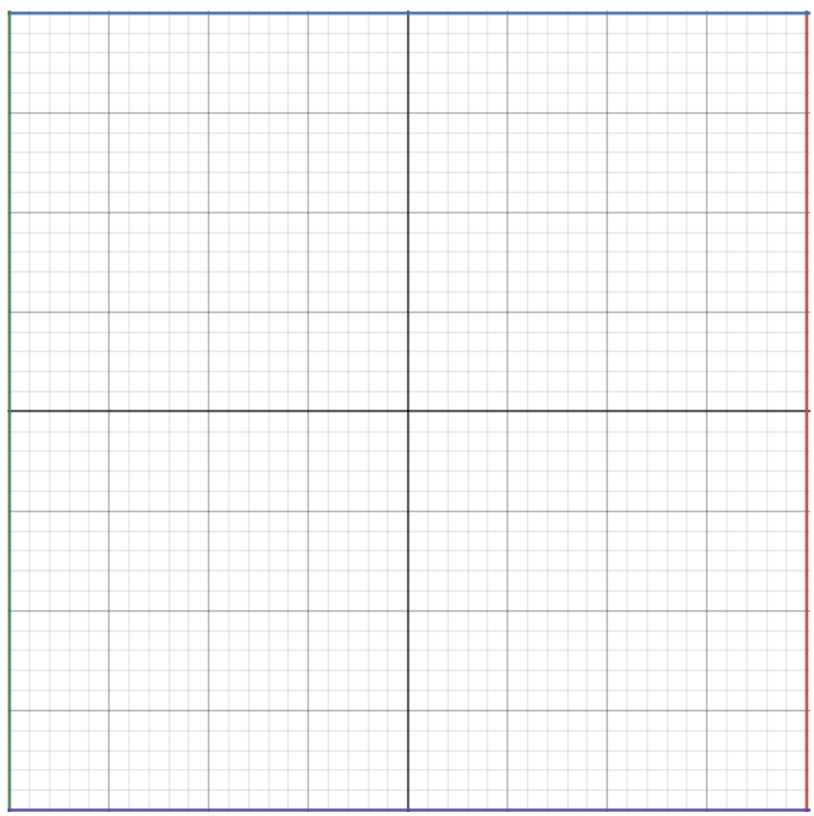
- **Label the 5 key points** for one complete cycle and **draw and label** the axis of the curve.
- **Label** all axes with proper scale and intervals.
- Extend your curve to the edges of the grid **to fill the space**.



33. Graph $f(x) = (2)^{\frac{-1}{3}x} + 1$ on the grid below by stating the mapping rule or listing the transformations.

For full marks, you must: **[6 marks]**

- List Transformations **OR** Mapping Rule Below.
- Include and **label** at least 3 points on the graph.
- Draw and **label** the asymptote.
- Label all axes with proper scale and intervals.



FINANCIAL FORMULAS

I. $A = P(1+i)^n$	II. $FV = \frac{R[(1+i)^n - 1]}{i}$	III. $PV = \frac{R[1-(1+i)^{-n}]}{i}$

34. For each of the scenarios, choose the best financial formula (I, II, or III) from above that would be used to solve the problem and write the corresponding number in the first column in the table below. In the second column, write which variable the problem requires you to find. **[3 marks]**

Problem	**Best Equation (I, II or III)**	**Variable being solved for**
George has saved \$40000 for his retirement. How much money can he withdraw each month, if the account earns 10%/a compounded monthly. *DO NOT SOLVE.*		
Elaine would like to donate \$50,000 to the local high school. She invests this amount into a high interest savings account that earns 7.2%/a compounded monthly, for 3 years. How much will the investment be worth at the end of the 3 years? *DO NOT SOLVE.*		
Seinfeld wants to buy an entertainment system in one year as a gift for his sister's wedding. He deposits \$225 each month into an account earning 3.5%/a compounded monthly. Will he have enough money to buy the gift in one year if it costs \$2499? *DO NOT SOLVE.*		

35. Sara wants to buy a \$2000 computer when she starts University in a year. Using one of the financial formulas stated above, determine how much money she will need to deposit into a savings account each month if the account earns 9%/a compounded monthly? **[5 marks]**